# Cheat Sheets

DTU 02477 Bayesian Machine Learning — All 12 weeks

## WEEK 1 Beta-Binomial & Bayesian Inference

### MODEL

$$p(y \mid N, \theta) = \binom{N}{y} \theta^y (1-\theta)^{N-y}$$

Prior: $\theta \sim \text{Beta}(a_0, b_0)$, with

$$p(\theta) = \frac{\theta^{a_0-1}(1-\theta)^{b_0-1}}{B(a_0, b_0)}, \quad B(a,b) = \frac{\Gamma(a}{\Gamma(a}$$

### CONJUGATE POSTERIOR UPDATE

$$p(\theta \mid y) = \text{Beta}(a_0 + y,\ b_0 + N - y)$$

### BETA DISTRIBUTION PROPERTIES

| Quantity | Formula |
|---|---|
| Mean | $\mu = \frac{a}{a+b}$ |
| Mode | $\hat{\theta} = \frac{a-1}{a+b-2}$ (for $a, b > 1$) |
| Variance | $\sigma^2 = \frac{ab}{(a+b)^2(a+b+1)}$ |

### POSTERIOR MEAN AS SHRINKAGE

$$\mathbb{E}[\theta \mid y] = (1-\lambda)\underbrace{\frac{a_0}{a_0+b_0}}_{\text{prior mean}} + \lambda\underbrace{\frac{y}{N}}_{\hat{\theta}_{\text{MLE}}}, \quad \lambda =$$

As $N \to \infty$: $\lambda \to 1$ (data dominates). As $N \to 0$: $\lambda \to 0$ (prior dominates).

### MLE & MODEL EVIDENCE

$$\hat{\theta}_{\text{MLE}} = \frac{y}{N}$$

$$p(y \mid N) = \binom{N}{y}\frac{B(a_0+y,\ b_0+N-y)}{B(a_0,b_0)}$$

### CREDIBILITY INTERVAL & MONTE CARLO

The $95\%$ credibility interval $[l, u]$ satisfies $P(l \leq \theta \leq u \mid y) = 0.95$.

Draw $\theta^{(s)} \sim p(\theta \mid y)$, then $\mathbb{E}[f(\theta) \mid y] \approx \frac{1}{S}\sum_{s=1}^{S} f(\theta^{(s)})$. Error $= O(1/\sqrt{S})$.

## WEEK 2 Bayesian Logistic Regression

### MODEL

$$p(y=1 \mid x, \alpha, \beta) = \sigma(\alpha + \beta x), \quad \sigma(z) =$$

Prior: $p(\alpha, \beta) \propto \exp\left(-\frac{\alpha^2}{2\sigma_\alpha^2} - \frac{\beta^2}{2\sigma_\beta^2}\right)$

### LOG JOINT

$$\log p(\alpha, \beta, \mathbf{y}) = \sum_n [y_n \log \sigma(f_n) + (1 - y_$$

#### GRID APPROXIMATION (3 STEPS)

1. Evaluate $\log p(\alpha_i, \beta_j, \mathbf{y})$ on a grid.
2. Subtract the maximum for stability, exponentiate: $\tilde{\pi}_{ij} = e^{\log p - \max}$.
3. Normalize: $\pi_{ij} = \tilde{\pi}_{ij} / \sum_{ij} \tilde{\pi}_{ij}$.

#### POSTERIOR PREDICTIVE

$$p(y^* = 1 \mid x^*, \mathbf{y}) = \sum_{i,j} \sigma(\alpha_i + \beta_j x^*)\, \pi_{ij}$$

#### KEY PROPERTIES

- Grid approximation is exact up to discretization; cost is $O(G^D)$ — exponential in dimension $D$.
- Bayesian predictions are more conservative (wider) than MAP predictions.

## WEEK 3 Bayesian Linear Regression

#### MODEL

$$p(\mathbf{y} \mid \mathbf{w}) = \mathcal{N}(\mathbf{y} \mid \mathbf{\Phi w},\, \beta^{-1}\mathbf{I}), \quad p(\mathbf{w}) = .$$

$\mathbf{\Phi} \in \mathbb{R}^{N \times D}$ is the design matrix with rows $\phi(x_n)^T$.

### CONJUGATE POSTERIOR

$$p(\mathbf{w} \mid \mathbf{y}) = \mathcal{N}(\mathbf{w} \mid \mathbf{m}, \mathbf{S})$$

$$\mathbf{S} = \left(\alpha \mathbf{I} + \beta \boldsymbol{\Phi}^T \boldsymbol{\Phi}\right)^{-1}, \qquad \mathbf{m} = \beta\, \mathbf{S}\, \boldsymbol{\Phi}^T \mathbf{y}$$

### POSTERIOR PREDICTIVE

For $f^* = \phi(x^*)^T \mathbf{w}$:

$$p(f^* \mid \mathbf{y}, x^*) = \mathcal{N}(f^* \mid \phi_*^T \mathbf{m},\ \phi_*^T \mathbf{S} \phi_*)$$

$$p(y^* \mid \mathbf{y}, x^*) = \mathcal{N}(y^* \mid \phi_*^T \mathbf{m},\ \phi_*^T \mathbf{S} \phi_* + \beta^-$$

### MLE (RIDGE)

$$\mathbf{w}_{\text{MLE}} = \left(\boldsymbol{\Phi}^T \boldsymbol{\Phi}\right)^{-1} \boldsymbol{\Phi}^T \mathbf{y}$$

### MARGINAL LIKELIHOOD (EVIDENCE)

$$p(\mathbf{y}) = \mathcal{N}\left(\mathbf{y} \mid \mathbf{0},\ \beta^{-1}\mathbf{I} + \alpha^{-1} \boldsymbol{\Phi}\boldsymbol{\Phi}^T\right)$$

> **Evidence approximation**: maximize $\log p(\mathbf{y})$ wrt. $\alpha, \beta$ (empirical Bayes / MLII). No cross-validation needed.

### KEY PROPERTIES

- Posterior mean $\mathbf{m}$ is a shrinkage estimator (pulled toward zero by $\alpha$).
- As $N \to \infty$, posterior concentrates around MLE.
- Posterior variance decreases as $N$ grows.

WEEK 4

# Laplace Approximation

### CORE IDEA

Approximate the posterior $p(\mathbf{w} \mid \mathbf{y})$ with a Gaussian centered at the MAP estimate:

$$q(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{w}_{\text{MAP}},\ \mathbf{A}^{-1})$$

$$\mathbf{A} = -\nabla^2 \log p(\mathbf{w}, \mathbf{y})\big|_{\mathbf{w}=\mathbf{w}_{\text{MAP}}}$$

($\mathbf{A}$ = negative Hessian of log joint = precision matrix.)

### STEPS

1. Find $\mathbf{w}_{\text{MAP}} = \arg\max_{\mathbf{w}} \log p(\mathbf{w}, \mathbf{y})$ via gradient-based optimization.
2. Evaluate $\mathbf{A} = -\nabla^2 \log p(\mathbf{w}, \mathbf{y})|_{\mathbf{w}_{\text{MAP}}}$.
3. Approximate posterior as $\mathcal{N}(\mathbf{w}_{\text{MAP}}, \mathbf{A}^{-1})$ .

### LOGISTIC REGRESSION: GRADIENT & HESSIAN

Let $f_n = \mathbf{w}^T\mathbf{x}_n$, $S = \text{diag}(\sigma(f_n)(1 - \sigma(f_n)))$:

$$\nabla_{\mathbf{w}} \log p = -\mathbf{X}^T(\boldsymbol{\sigma} - \mathbf{y}) - \alpha\mathbf{w}$$

$$\nabla^2_{\mathbf{w}} \log p = -\mathbf{X}^T S\mathbf{X} - \alpha\mathbf{I} \quad \Rightarrow \quad \mathbf{A} = \mathbf{X}$$

### PREDICTIVE APPROXIMATIONS

Let $f^* = \mathbf{w}^T\phi_*$, so $f^* \sim \mathcal{N}(m^*, v^*)$ with $m^* = \mathbf{w}_{\text{MAP}}^T\phi_*$, $v^* = \phi_*^T\mathbf{A}^{-1}\phi_*$.

| Method | Formula |
|---|---|
| Plug-in | $p(y^* = 1) \approx \sigma(m^*)$ |
| Probit | $p(y^* = 1) \approx \sigma\left(\frac{m^*}{\sqrt{1 + \pi v^*/8}}\right)$ |
| MC | $p(y^* = 1) \approx \frac{1}{S}\sum_s \sigma(f^{*(s)})$, $f^{*(s)} \sim \mathcal{N}(m^*, v^*)$ |

### GENERATIVE VS DISCRIMINATIVE

- **Generative**: models $p(y, x) = p(x \mid y)p(y)$; class-conditional + prior.
- **Discriminative**: models $p(y \mid x)$ directly; logistic regression, neural networks.

## WEEK 5 Gaussian Processes

### GP PRIOR

A GP is defined by a mean function $m(x)$ and kernel $k(x, x')$:

$$f \sim \mathcal{GP}(m(\cdot),\, k(\cdot, \cdot))$$

Typically $m(x) = 0$.

### COMMON KERNELS

**Squared Exponential (SE):**

$$k_{\text{SE}}(x, x') = \sigma_f^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right)$$

**Matérn 3/2:**

$$k_{\text{M32}}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{3}\,r}{\ell}\right) \exp\left(-\frac{\sqrt{3}\,r}{\ell}\right)$$

Hyperparameters: length-scale $\ell$ (smoothness), signal variance $\sigma_f^2$ (amplitude), noise variance $\sigma_n^2$.

### GP REGRESSION POSTERIOR

Given $\mathbf{y} = f(\mathbf{X}) + \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2\mathbf{I})$, let $\mathbf{C} = K(\mathbf{X}, \mathbf{X}) + \sigma_n^2\mathbf{I}$:

$$\boldsymbol{\mu}_{f^*} = K(\mathbf{X}^*, \mathbf{X})\,\mathbf{C}^{-1}\mathbf{y}$$

$$\boldsymbol{\Sigma}_{f^*} = K(\mathbf{X}^*, \mathbf{X}^*) - K(\mathbf{X}^*, \mathbf{X})\,\mathbf{C}^{-1}\,K(\mathbf{X}$$

Predictive: $p(y^* \mid \mathbf{y}, x^*) = \mathcal{N}(y^* \mid \mu_{f^*}, \Sigma_{f^*} + \sigma_n^2)$.

### LOG MARGINAL LIKELIHOOD (MLII)

$$\log p(\mathbf{y}) = -\frac{1}{2}\mathbf{y}^T\mathbf{C}^{-1}\mathbf{y} - \frac{1}{2}\log|\mathbf{C}| - \frac{N}{2}$$

Cholesky: $\mathbf{L} = \text{chol}(\mathbf{C})$, so $\log|\mathbf{C}| = 2\sum_i \log L_{ii}$.

**Hyperparameter optimization**: maximize $\log p(\mathbf{y})$ wrt. $\ell, \sigma_f^2, \sigma_n^2$ via gradient ascent.

### KEY PROPERTIES

- Posterior mean = kernel-weighted interpolation.
- Posterior variance = 0 at training points (noise-free).
- Short $\ell$: wiggly. Long $\ell$: smooth.

WEEK 6 **GP Classification**

### MODEL

$$p(y_n = 1 \mid f_n) = \sigma(f_n), \quad f \sim \mathcal{GP}(0, k(\cdot,$$

The posterior $p(\mathbf{f} \mid \mathbf{y})$ is **non-Gaussian** — no conjugacy.

### LAPLACE APPROXIMATION FOR LATENT GP

Find $\hat{\mathbf{f}} = \arg\max_{\mathbf{f}} \log p(\mathbf{f} \mid \mathbf{y})$ and approximate $q(\mathbf{f}) = \mathcal{N}(\hat{\mathbf{f}},\ (K^{-1} + \Lambda)^{-1})$, where $\Lambda = \mathrm{diag}(\lambda_n)$, $\lambda_n = \sigma(f_n)(1 - \sigma(f_n))$.

**Gradient:** $\mathbf{g} = \mathbf{y} - \boldsymbol{\sigma}(\mathbf{f}) - K^{-1}\mathbf{f}$

**Newton update:** $\mathbf{f}^{\text{new}} = \mathbf{f} + (K^{-1} + \Lambda)^{-1}\mathbf{g}$

**Numerically stable covariance** (Woodbury): $\mathbf{S} = K - K\Lambda^{1/2}\mathbf{B}^{-1}\Lambda^{1/2}K$, $\mathbf{B} = \mathbf{I} + \Lambda^{1/2}K\Lambda^{1/2}$

### PREDICTIVE DISTRIBUTION (3-STEP PIPELINE)

**Step 1** — posterior mean:

$$\mu_{f^*} = k(x^*, \mathbf{X})\, K^{-1}\hat{\mathbf{f}}$$

**Step 2** — posterior variance:

$$\sigma^2_{f^*} = k(x^*, x^*) - k(x^*, \mathbf{X})\, (K + \Lambda^{-1})^{-1}\, k$$

**Step 3** — probit approximation:

$$p(y^* = 1 \mid \mathbf{y}, x^*) \approx \sigma\left(\frac{\mu_{f^*}}{\sqrt{1 + \pi\sigma_{f^*}^2/8}}\right)$$

Or MC: $p(y^* = 1) \approx \frac{1}{S}\sum_s \sigma(f^{*(s)})$, $f^{*(s)} \sim \mathcal{N}(\mu_{f^*}, \sigma_{f^*}^2)$.

WEEK 7

## Multi-class Classification

### SOFTMAX MODEL

$$\mathbf{f}_n = W\phi(x_n) \in \mathbb{R}^C, \quad p(y_n = c \mid \mathbf{f}_n) = \text{sc}$$

Prior: $p(W) = \mathcal{N}(\text{vec}(W) \mid \mathbf{0}, \alpha^{-1}\mathbf{I})$

### LOG JOINT & LAPLACE APPROXIMATION

$$\log p(W, \mathbf{y}) = \sum_n \log \text{softmax}(\mathbf{f}_n)_{y_n} - \frac{\alpha}{2}|$$

MAP: $W_{\text{MAP}} = \arg\max_W \log p(W, \mathbf{y})$

$\mathbf{A} = -\nabla^2 \log p(W, \mathbf{y})|_{W_{\text{MAP}}}$ (JAX Hessian), $q(W) = \mathcal{N}(W_{\text{MAP}}, \mathbf{A}^{-1})$.

### MC POSTERIOR PREDICTIVE

Sample $W^{(s)} \sim q(W)$:

$$\boldsymbol{\pi}^* \approx \frac{1}{S}\sum_{s=1}^{S} \text{softmax}(W^{(s)}\phi(x^*))$$

### UNCERTAINTY DECOMPOSITION

$$H_{\text{total}} = -\sum_c \pi_c^* \log \pi_c^*$$

$$H_{\text{aleatoric}} = \frac{1}{S}\sum_s H\Big[\text{Cat}(\text{softmax}(W^{(s)}\phi$$

$$H_{\text{epistemic}} = H_{\text{total}} - H_{\text{aleatoric}} \geq 0$$

High $H_{\text{epistemic}}$ → out-of-distribution. High $H_{\text{aleatoric}}$ → inherently ambiguous.

### BAYESIAN DECISION THEORY & ECE

Optimal action: $\hat{y}^* = \arg\max_a \sum_c \pi_c^* U(c,a)$, where $U(c,a)$ is the utility of action $a$ when true class is $c$.

$$\text{ECE} = \sum_b \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$$

## WEEK 8 MCMC (Metropolis-Hastings)

### METROPOLIS-HASTINGS ALGORITHM

Goal: draw $\theta^{(1)}, \ldots, \theta^{(K)} \sim p(\theta \mid \mathbf{y}) \propto p_t(\theta)$.

1. Propose $\theta^* \sim q(\theta^* \mid \theta^{(k-1)})$ (e.g. random walk $\theta^* = \theta^{(k-1)} + \epsilon, \epsilon \sim \mathcal{N}(0, \eta^2 I)$).
2. Compute log acceptance ratio (symmetric proposal): $\log r_k = \log p_t(\theta^*) - \log p_t(\theta^{(k-1)})$
3. Set $\theta^{(k)} = \theta^*$ with probability $\min(1, e^{\log r_k})$, else $\theta^{(k)} = \theta^{(k-1)}$.

Asymmetric proposal: add $\log q(\theta^{(k-1)} \mid$

$\theta^*) - \log q(\theta^* \mid \theta^{(k-1)})$ to $\log r_k$.

### BAYESIAN POISSON REGRESSION WITH HYPERPRIOR

$y_n \sim \text{Poisson}(\exp(\mathbf{w}^T \mathbf{x}_n)), \quad p(\mathbf{w} \mid \kappa) =$

Log joint: $\sum_n [y_n(\mathbf{w}^T \mathbf{x}_n) - e^{\mathbf{w}^T \mathbf{x}_n}] - \frac{\|\mathbf{w}\|^2}{2\kappa^2} + \log p(\kappa)$

### IMPORTANCE SAMPLING

$$\mathbb{E}_p[f(\theta)] \approx \frac{\sum_i w^{(i)} f(\theta^{(i)})}{\sum_i w^{(i)}}, \quad w^{(i)} = \frac{p(\theta^{(i)}}{q(\theta^{(i)}}$$

### ANCESTRAL SAMPLING

Draw from joint $p(\theta_1, \theta_2, \dots)$ in topological order: $\theta_1 \sim p(\theta_1)$, then $\theta_2 \sim p(\theta_2 \mid \theta_1)$, etc.

### PRACTICAL NOTES

- **Burn-in**: discard early samples until chain mixes.
- **Step size** $\eta$: too large → high rejection; too small → slow mixing. Target ~23–65% acceptance.
- **Thinning**: keep every $k$-th sample to reduce autocorrelation.

WEEK 9

# Hamiltonian Monte Carlo (HMC)

### HAMILTONIAN SYSTEM

Introduce momentum $\nu \sim \mathcal{N}(0, \mathbf{I})$ and simulate dynamics to get distant proposals with high acceptance:

$$H(\theta, \nu) = E(\theta) + K(\nu), \quad E(\theta) = -\log$$

Dynamics: $\dot{\theta} = \nu, \dot{\nu} = -\nabla_\theta E(\theta)$.

### LEAPFROG INTEGRATOR ($L$ STEPS, STEP SIZE $H$)

$$\nu_{j+1/2} = \nu_j - \frac{\eta}{2} \nabla_\theta E(\theta_j)$$

$$\theta_{j+1} = \theta_j + \eta\, \nu_{j+1/2}$$

$$\nu_{j+1} = \nu_{j+1/2} - \frac{\eta}{2} \nabla_\theta E(\theta_{j+1})$$

### HMC ACCEPTANCE & SAMPLING

Accept $(\theta^*, \nu^*)$ with probability:

$$a_k = \min\left(1,\ e^{-H(\theta^*, \nu^*) + H(\theta^{(k-1)}, \nu^{(k-1)})}\right)$$

Discard $\nu^*$ after; sample fresh $\nu \sim \mathcal{N}(0, I)$ each iteration.

### CONVERGENCE DIAGNOSTICS

**R-hat** (run $M$ chains of length $S$):

$$\hat{R}^2 = \frac{S-1}{S} + \frac{1}{S}\frac{B}{W}$$

$B$ = between-chain variance, $W$ = within-chain variance. Want $\hat{R} < 1.01$.

**ESS:**

$$S_{\text{eff}} = \frac{S}{1 + 2\sum_{t=1}^{\infty} \hat{\rho}_t}$$

$\hat{\rho}_t$ = autocorrelation at lag $t$. Want $S_{\text{eff}} \geq 100$ per chain.

### HETEROSCEDASTIC REGRESSION & HMC VS MH

Positivity of variance via softplus: $\sigma_n^2 = \text{softplus}(g(\mathbf{x}_n)) = \log(1 + e^{g(\mathbf{x}_n)})$

| | MH | HMC |
|---|---|---|
| Proposals | random walk | gradient-guided |
| Acceptance | often low in high-D | near 1 with good tuning |
| Autocorrelation | high | low |
| Requires | $\log p$ | $\log p$ + gradient |

## WEEK 10 Variational Inference & CAVI

### ELBO DECOMPOSITION

$$\log p(\mathbf{X}) = \underbrace{\mathcal{L}[q]}_{\text{ELBO}} + \text{KL}[q(\mathbf{Z}) \| p(\mathbf{Z} \mid \mathbf{X})] \geq$$

$\mathcal{L}[q] = \mathbb{E}_q[\log p(\mathbf{X}, \mathbf{Z})] - \mathbb{E}_q[\log q(\mathbf{Z})] = \mathbb{E}$

$\mathrm{KL}[q||p] = \mathbb{E}_q[\log q - \log p] \geq 0$, equality iff $q = p$.

### MEAN-FIELD FAMILY & CAVI

$$q(\mathbf{Z}) = \prod_i q_i(Z_i)$$

Optimal factor (all others fixed):

$$\log q_j^*(Z_j) = \mathbb{E}_{-j}[\log p(\mathbf{X}, \mathbf{Z})] + \text{const}$$

Cycle through all factors until ELBO converges.

### GAUSSIAN-GAMMA CAVI UPDATES

Model: $x_n \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^{-1})$, prior: $\mu \mid \tau \sim \mathcal{N}(\mu_0, (\lambda_0 \tau)^{-1})$, $\tau \sim \mathrm{Gamma}(a_0, b_0)$.

$$q^*(\mu) = \mathcal{N}(\mu \mid m, v), \quad v = \frac{1}{\mathbb{E}[\tau](N + \lambda_0}$$

$$q^*(\tau) = \mathrm{Gamma}(\tau \mid a, b), \quad a = a_0 + \frac{N}{}$$

### BAYESIAN GMM KEY QUANTITIES

**Responsibilities:** $r_{nk} = \dfrac{\pi_k \mathcal{N}(x_n \mid \mu_k, \Lambda_k^{-1})}{\sum_j \pi_j \mathcal{N}(x_n \mid \mu_j, \Lambda_j^{-1})}$

**Effective counts:** $N_k = \sum_n r_{nk}$

**Posterior cluster mean** (shrinkage):

$$m_k = (1 - \rho_k) m_0 + \rho_k \bar{x}_k, \quad \rho_k = \frac{N_k}{\beta_k},$$

Prior: $\boldsymbol{\pi} \sim \mathrm{Dir}(\alpha_0 \mathbf{1})$, $\mu_k \sim$

$\mathcal{N}(m_0, \beta_0^{-1}\Lambda_k^{-1})$, $\Lambda_k \sim \text{Wishart}(W_0, \nu_0)$.

#### MEAN-FIELD UNDERESTIMATES VARIANCE

Optimal diagonal covariance: $\hat{V}_{ii} = 1/(V^{-1})_{ii} \leq V_{ii}$ whenever off-diagonal correlations are nonzero.

WEEK 11

## Black-Box Variational Inference (BBVI)

#### ELBO & MEAN-FIELD GAUSSIAN ENTROPY

$$\mathcal{L}[q] = \underbrace{\mathbb{E}_{q(\mathbf{w})}[\log p(\mathbf{y}, \mathbf{w})]}_{\text{expected log joint}} + \underbrace{\mathcal{H}[q]}_{\text{entropy}}$$

Mean-field: $q(\mathbf{w}) = \prod_{i=1}^{D} \mathcal{N}(w_i \mid m_i, v_i)$, variational parameters $\lambda = \{\mathbf{m}, \mathbf{v}\}$.

Analytical entropy: $\mathcal{H}[q] = \frac{1}{2} \sum_{i=1}^{D} \log(2e\pi v_i)$

#### BBVI MONTE CARLO ELBO

$$\mathcal{L}[q] \approx \frac{1}{S} \sum_{s=1}^{S} \log p(\mathbf{y}, \mathbf{w}^{(s)}) + \mathcal{H}[q], \quad \mathbf{w}^{(s}$$

#### REPARAMETRIZATION TRICK

Replace $\mathbf{w} \sim \mathcal{N}(\mathbf{m}, \text{diag}(\mathbf{v}))$ with:

$$\mathbf{w} = \mathbf{m} + \mathbf{s} \circ \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad s_i = \sqrt{\imath}$$

Moves $\lambda$-dependence outside the expectation, enabling gradient computation via autodiff:

$$\nabla_\lambda \mathbb{E}_{q_\lambda}[\log p(\mathbf{y}, \mathbf{w})] \approx \frac{1}{S} \sum_s \nabla_\lambda \log p(\mathbf{y}, \mathbf{w}$$

### MINIBATCHING

For large $N$, approximate the log-likelihood term using a batch $\mathcal{M}$ of size $M \ll N$:

$$\mathbb{E}_q[\log p(\mathbf{y} \mid \mathbf{w})] \approx \frac{N}{M} \sum_{n \in \mathcal{M}} \mathbb{E}_q[\log p(y_n \mid \mathbf{w}$$

Unbiased estimator when $\mathcal{M}$ is resampled each step.

### ANALYTICAL ELBO FOR LINEAR GAUSSIAN

$$\mathcal{L} = \underbrace{\sum_n \log \mathcal{N}(y_n \mid \mathbf{m}^T \mathbf{x}_n, \sigma^2) - \frac{1}{2\sigma^2} \sum_{i,n}}_{\text{expected log likelihood}}$$

### LIKELIHOOD VARIANTS FOR BINARY CLASSIFICATION

| Model | $p(y = 1 \mid f)$ | Property |
|---|---|---|
| Sigmoid | $\sigma(f) = (1 + e^{-f})^{-1}$ | Standard logistic |
| Probit | $\Phi(f)$ (Gaussian CDF) | Heavier tails |
| Robust | $(1 - \epsilon)\sigma(f) + \epsilon\sigma(-f)$ | Tolerates mislabelled data |

Optimize $\lambda = \{\mathbf{m}, \log \mathbf{v}\}$ (unconstrained, ensures $v_i > 0$) with **Adam**.

## WEEK 12 Bayesian Deep Learning

### HETEROSCEDASTIC NEURAL NETWORK

Two-output network: $f(\mathbf{x} \mid \mathbf{w}) \in \mathbb{R}^2$

$\mu_{\mathbf{w}}(\mathbf{x}) = f_1(\mathbf{x} \mid \mathbf{w}), \quad \sigma^2_{\mathbf{w}}(\mathbf{x}) = e^{f_2(\mathbf{x}|\mathbf{w})} >$

$p(y_n \mid \mathbf{w}, \mathbf{x}_n) = \mathcal{N}(y_n \mid \mu_{\mathbf{w}}(\mathbf{x}_n),\ \sigma^2_{\mathbf{w}}(\mathbf{x}_n)),$

### MAP INFERENCE

$$\mathbf{w}_{\text{MAP}} = \arg\max_{\mathbf{w}} \left[ \sum_n \log p(y_n \mid \mathbf{w}, \mathbf{x}_n) - \right.$$

Equivalent to L2-regularized MLE. Plugin predictive: $p(y^* \mid \mathbf{w}_{\text{MAP}}, \mathbf{x}^*)$.

Approximate posterior predictive: $p(y^* \mid \mathbf{y}, \mathbf{x}^*) \approx \frac{1}{S}\sum_{s=1}^{S} p(y^* \mid \mathbf{w}^{(s)}, \mathbf{x}^*)$, $\mathbf{w}^{(s)} \sim q(\mathbf{w})$.

### LAST-LAYER LAPLACE APPROXIMATION (LLLA)

Apply Laplace only to last-layer parameters $\mathbf{w}_L = \{\mathbf{W}_2, \mathbf{b}_2\}$:

1. Compute $\mathbf{w}_{\text{MAP}}$ (full network via gradient descent).
2. Compute features: $\mathbf{z} = f_{1:L-1}(\mathbf{X} \mid \mathbf{w}_{1:L-1,\text{MAP}})$.

3. Hessian wrt. last layer only: $\mathbf{H} = \mathbf{H}_{\text{loglik}} + \alpha\mathbf{I}$
4. Posterior: $q(\mathbf{w}_L) = \mathcal{N}(\mathbf{w}_{L,\text{MAP}},\ \mathbf{H}^{-1})$.
5. Predict: sample $\mathbf{w}_L^{(s)} \sim q$, compute $\mathbf{f}^{(s)} = \mathbf{W}_2^{(s)}\mathbf{z} + \mathbf{b}_2^{(s)}$.

> **Benefits**: cheap (small Hessian), post-hoc (no retraining), captures uncertainty in final mapping.

### DEEP ENSEMBLES (DE)

Train $S$ networks from different initializations: $q_{\text{DE}}(\mathbf{w}) = \frac{1}{S}\sum_i \delta(\mathbf{w} - \mathbf{w}^{(i)})$

Predictive (Gaussian mixture):

$$p(y^* \mid \mathbf{y}, \mathbf{x}^*) \approx \frac{1}{S}\sum_{i=1}^{S} \mathcal{N}(y^* \mid f_1^{(i)}(\mathbf{x}^*),\ e^{f_2^{(i}}$$

MAP = degenerate ensemble with $S = 1$.

### LOG PREDICTIVE DENSITY (LPD)

$$\text{LPD}(y^*) \approx \log\left[\frac{1}{S}\sum_s p(y^* \mid \mathbf{w}^{(s)}, \mathbf{x}^*)\right]$$

Average LPD $= \frac{1}{N_{\text{test}}}\sum_n \text{LPD}(y_n^*)$. Higher is better.

### METHOD COMPARISON

| Method | Uncertainty | Cost | OOD quality |
|---|---|---|---|
| MAP | None (point) | Low | Poor |
| LLLA | Epistemic | Low | Moderat |

| | | | |
|---|---|---|---|
| | (last layer) | (post-hoc) | |
| Deep Ensembles | Epistemic (functional) | High ( $S\times$) | Good |
| Full Laplace | Epistemic (full) | Very high | Good |
| HMC | Full posterior | Highest | Best |

### UNCERTAINTY SOURCES

- **Aleatoric** ($\sigma^2_{\mathbf{w}}(\mathbf{x})$): data noise, irreducible, from heteroscedastic likelihood.
- **Epistemic** (spread of $q(\mathbf{w})$): model uncertainty, reducible with more data, from posterior.